

ELECTROLYTE RELATIONSHIP TABLE

How the major electrolytes interact and what those relationships mean clinically

↑ increases ↓ decreases ↔ closely related • Opposite = often inverse • Together = commonly same clinical direction

RELATIONSHIP	WHAT HAPPENS?	WHY / PHYSIOLOGY	WHAT YOU MAY SEE	NURSING / EXAM CONNECTION
Ca ↔ phosphate OPPOSITE • often	↓ Ca ↔ ↑ phosphate ↑ Ca ↔ ↓ phosphate	Phosphate binds Ca. PTH raises serum Ca and promotes phosphate loss in urine.	High phosphate may look like low Ca: tingling, spasms, hyperreflexia , tetany.	Trousseau : BP cuff → hand spasm. Chvostek : face tap → twitch. Severe: SEIZURE / LARYNGOSPASM risk. Assess neuro + ECG .
Mg ↔ Ca NEUROMUSCULAR / REFLEXES	↓ Mg + ↓ Ca → excitability ↑ ↑ Mg → CNS / muscle depression	Both stabilize nerve/muscle activity. Severe Mg lack can impair Ca regulation.	Low: tremors, cramps, hyperreflexia, tetany. High: ↓ DTRs, lethargy, weakness, low BP/HR, RESPIRATORY DEPRESSION .	Check DTRs , breathing + ECG . LOW = too excitable. HIGH = too relaxed/depressed.
K ↔ acid-base HEART / MUSCLE	ACIDOSIS → K tends OUT → serum K ↑ ALKALOSIS → K tends IN → serum K ↓	H ⁺ and K shift across cell membranes as the body responds to pH.	Either K extreme → weakness + DYRHYTHMIA risk. Alkalosis may ↓ ionized Ca → cramps/tetany.	WATCH K + ECG . A tendency, not a rule. Cause matters; GI loss can still lower K.
K ↔ insulin / glucose EXAM 1 • DIABETES	INSULIN → GLUCOSE + K INTO CELLS → serum K can fall	Insulin stimulates K uptake. DKA causes fluid/electrolyte loss; treatment can lower serum K further.	K may drop during insulin therapy → weakness + ECG/rhythm changes.	Trend K before/during insulin; monitor rhythm as indicated. SERUM K ≠ TOTAL-BODY K .
Cl ↔ bicarbonate ACID-BASE	Vomiting / NG : lose H ⁺ + Cl → ↓ Cl → alkalosis; K often ↓. Diarrhea : lose HCO ₃ → acidosis; Cl may ↑.	Gastric fluid contains acid + Cl. Intestinal loss removes HCO ₃ . Read Cl and HCO ₃ together.	Vomiting/NG: weakness, cramps, alkalosis. Diarrhea: fluid loss, acidosis; deep/fast breathing may occur.	READ TOGETHER: Cl + HCO₃ + K + FLUID STATUS .
Na ↔ water WATER ↔ BRAIN	Water excess vs Na → dilution → Na ↓ Water deficit vs Na → concentration → Na ↑	Serum Na depends strongly on water relative to Na. Water shifts toward higher osmolality.	Brain cells swell or shrink. Significant changes → confusion, LOC change, SEIZURES .	SODIUM = NEURO + FLUID . Check neuro, I&O; weight. AVOID RAPID Na CORRECTION → neurologic injury.
Na ↔ hormones ADH / ALDOSTERONE	↑ ADH : retain water → dilute Na ↓ ↓ / ineffective ADH (DI) : water loss → Na may ↑ Aldosterone : retain Na/water; excrete K	ADH controls water retention. Aldosterone promotes Na retention and K loss.	SIADH → dilutional hyponatremia. DI → high urine output, water loss, hypernatremia risk. Excess aldosterone → K may fall.	Compare Na + K + I&O; + urine output + BP + daily weight .
Phosphate ↔ energy ATP / MUSCLE / RESPIRATORY	↓ phosphate → ↓ cellular energy / ATP → muscle function ↓	Phosphate is needed for cellular energy production and muscle function.	General weakness, weak respiratory muscles, poor activity tolerance, ↓ cardiac performance, fall risk.	LOW PHOSPHATE = THINK WEAK . Check breathing strength, mobility, fall/cardiac risk + RENAL FUNCTION .
Mg ↔ K CORRECTION CLUE	↓ Mg → kidney K wasting → low K may persist	Mg deficiency makes it harder for the kidneys to retain K.	K stays low despite replacement; combined deficits increase rhythm risk.	K not improving? CHECK Mg . Monitor ECG + kidney function .
Kidneys ↔ all PUT IT TOGETHER	Impaired regulation / excretion → several abnormalities at once	Kidneys regulate electrolytes, acid-base and fluid. K, Mg and phosphate may accumulate when excretion falls.	Rhythm, reflex, weakness, fluid and pH changes may occur together.	THINK: KIDNEYS + GI LOSSES + MEDS + FLUIDS . Review BUN/Cr, urine output, I&O; , meds, orders + trends.

PATTERNS TO RECOGNIZE ON AN EXAM

ACIDOSIS K tends OUT → serum K ↑ Check actual K + ECG	ALKALOSIS K tends IN → K ↓ ionized Ca ↓ → tetany risk	Ca / PHOSPHATE Often opposite High phosphate can look like low Ca	LOW Mg + LOW Ca Excitability ↑ Cramps / hyperreflexia / tetany
VOMITING / NG H ⁺ + Cl lost → alkalosis K often ↓	DIARRHEA HCO ₃ lost → acidosis Cl may ↑; K may ↓ from GI loss	INSULIN / DKA Insulin moves K into cells Serum K may fall; trend closely	RENAL FAILURE K / Mg / phosphate may build up Check creatinine + urine output